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compact1, compact2, compact3

java.lang.reflect

Class Modifier

java.lang.Object
    java.lang.reflect.Modifier

```
public class Modifier
extends Object
```

The Modifier class provides `static` methods and constants to decode class and member access modifiers. The sets of modifiers are represented as integers with distinct bit positions representing different modifiers. The values for the constants representing the modifiers are taken from the tables in sections 4.1, 4.4, 4.5, and 4.7 of *The Java™ Virtual Machine Specification*.

See Also:

`Class.getModifiers()`, `Member.getModifiers()`

Field Summary

Fields

Modifier and Type	Field	Description
`static int`	**`ABSTRACT`**	The `int` value representing the `abstract` modifier.
`static int`	**`FINAL`**	The `int` value representing the `final` modifier.
`static int`	**`INTERFACE`**	The `int` value representing the `interface` modifier.
`static int`	**`NATIVE`**	

		The int value representing the native modifier.
static int	PRIVATE	The int value representing the private modifier.
static int	PROTECTED	The int value representing the protected modifier.
static int	PUBLIC	The int value representing the public modifier.
static int	STATIC	The int value representing the static modifier.
static int	STRICT	The int value representing the strictfp modifier.
static int	SYNCHRONIZED	The int value representing the synchronized modifier.
static int	TRANSIENT	The int value representing the transient modifier.
static int	VOLATILE	The int value representing the volatile modifier.

Constructor Summary

Constructors

Constructor	Description
Modifier()	

Method Summary

Concrete Methods

Modifier and Type	Method	Description
static int	classModifiers()	Return an int value OR-ing together the source language modifiers that can be applied to a class.
static int	constructorModifiers()	

		Return an `int` value OR-ing together the source language modifiers that can be applied to a constructor.
`static int`	**`fieldModifiers`**`()`	Return an `int` value OR-ing together the source language modifiers that can be applied to a field.
`static int`	**`interfaceModifiers`**`()`	Return an `int` value OR-ing together the source language modifiers that can be applied to an interface.
`static boolean`	**`isAbstract`**`(int mod)`	Return `true` if the integer argument includes the `abstract` modifier, `false` otherwise.
`static boolean`	**`isFinal`**`(int mod)`	Return `true` if the integer argument includes the `final` modifier, `false` otherwise.
`static boolean`	**`isInterface`**`(int mod)`	Return `true` if the integer argument includes the `interface` modifier, `false` otherwise.
`static boolean`	**`isNative`**`(int mod)`	Return `true` if the integer argument includes the `native` modifier, `false` otherwise.
`static boolean`	**`isPrivate`**`(int mod)`	Return `true` if the integer argument includes the `private` modifier, `false` otherwise.
`static boolean`	**`isProtected`**`(int mod)`	Return `true` if the integer argument includes the `protected` modifier, `false` otherwise.
`static boolean`	**`isPublic`**`(int mod)`	Return `true` if the integer argument includes the

		public modifier, false otherwise.
static boolean	**isStatic**(int mod)	Return true if the integer argument includes the static modifier, false otherwise.
static boolean	**isStrict**(int mod)	Return true if the integer argument includes the strictfp modifier, false otherwise.
static boolean	**isSynchronized**(int mod)	Return true if the integer argument includes the synchronized modifier, false otherwise.
static boolean	**isTransient**(int mod)	Return true if the integer argument includes the transient modifier, false otherwise.
static boolean	**isVolatile**(int mod)	Return true if the integer argument includes the volatile modifier, false otherwise.
static int	**methodModifiers**()	Return an int value OR-ing together the source language modifiers that can be applied to a method.
static int	**parameterModifiers**()	Return an int value OR-ing together the source language modifiers that can be applied to a parameter.
static **String**	**toString**(int mod)	Return a string describing the access modifier flags in the specified modifier.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Detail

PUBLIC

```
public static final int PUBLIC
```

The int value representing the public modifier.

See Also:

Constant Field Values

PRIVATE

```
public static final int PRIVATE
```

The int value representing the private modifier.

See Also:

Constant Field Values

PROTECTED

```
public static final int PROTECTED
```

The int value representing the protected modifier.

See Also:

Constant Field Values

STATIC

```
public static final int STATIC
```

The int value representing the static modifier.

See Also:

Constant Field Values

FINAL

```
public static final int FINAL
```

The `int` value representing the `final` modifier.

See Also:

Constant Field Values

SYNCHRONIZED

```
public static final int SYNCHRONIZED
```

The `int` value representing the `synchronized` modifier.

See Also:

Constant Field Values

VOLATILE

```
public static final int VOLATILE
```

The `int` value representing the `volatile` modifier.

See Also:

Constant Field Values

TRANSIENT

```
public static final int TRANSIENT
```

The `int` value representing the `transient` modifier.

See Also:

Constant Field Values

NATIVE

```
public static final int NATIVE
```

The `int` value representing the `native` modifier.

See Also:

Constant Field Values

INTERFACE

```
public static final int INTERFACE
```

The `int` value representing the `interface` modifier.

See Also:

Constant Field Values

ABSTRACT

```
public static final int ABSTRACT
```

The `int` value representing the `abstract` modifier.

See Also:

Constant Field Values

STRICT

```
public static final int STRICT
```

The `int` value representing the `strictfp` modifier.

See Also:

Constant Field Values

Constructor Detail

Modifier

```
public Modifier()
```

Method Detail

isPublic

```
public static boolean isPublic(int mod)
```

Return `true` if the integer argument includes the `public` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the public modifier; false otherwise.`

isPrivate

```
public static boolean isPrivate(int mod)
```

Return `true` if the integer argument includes the `private` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the private modifier; false otherwise.`

isProtected

```
public static boolean isProtected(int mod)
```

Return `true` if the integer argument includes the `protected` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the protected modifier; false otherwise.`

isStatic

```
public static boolean isStatic(int mod)
```

Return `true` if the integer argument includes the `static` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the static modifier; false otherwise.`

isFinal

```
public static boolean isFinal(int mod)
```

Return `true` if the integer argument includes the `final` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the final modifier; false otherwise.`

isSynchronized

```
public static boolean isSynchronized(int mod)
```

Return `true` if the integer argument includes the `synchronized` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the synchronized modifier; false otherwise.`

isVolatile

```
public static boolean isVolatile(int mod)
```

Return `true` if the integer argument includes the `volatile` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the volatile modifier; false otherwise.`

isTransient

```
public static boolean isTransient(int mod)
```

Return `true` if the integer argument includes the `transient` modifier, `false` otherwise.

Parameters:

mod - a set of modifiers

Returns:

true if mod includes the transient modifier; false otherwise.

isNative

```
public static boolean isNative(int mod)
```

Return `true` if the integer argument includes the `native` modifier, `false` otherwise.

Parameters:

mod - a set of modifiers

Returns:

true if mod includes the native modifier; false otherwise.

isInterface

```
public static boolean isInterface(int mod)
```

Return `true` if the integer argument includes the `interface` modifier, `false` otherwise.

Parameters:

mod - a set of modifiers

Returns:

true if mod includes the interface modifier; false otherwise.

isAbstract

```
public static boolean isAbstract(int mod)
```

Return `true` if the integer argument includes the `abstract` modifier, `false` otherwise.

Parameters:

mod - a set of modifiers

Returns:

true if mod includes the abstract modifier; false otherwise.

isStrict

```
public static boolean isStrict(int mod)
```

Return `true` if the integer argument includes the `strictfp` modifier, `false` otherwise.

Parameters:

`mod - a set of modifiers`

Returns:

`true if mod includes the strictfp modifier; false otherwise.`

toString

```
public static String toString(int mod)
```

Return a string describing the access modifier flags in the specified modifier. For example:

```
public final synchronized strictfp
```

The modifier names are returned in an order consistent with the suggested modifier orderings given in sections 8.1.1, 8.3.1, 8.4.3, 8.8.3, and 9.1.1 of *The Java™ Language Specification*. The full modifier ordering used by this method is:

```
public protected private abstract static final transient
volatile synchronized native strictfp interface
```

The `interface` modifier discussed in this class is not a true modifier in the Java language and it appears after all other modifiers listed by this method. This method may return a string of modifiers that are not valid modifiers of a Java entity; in other words, no checking is done on the possible validity of the combination of modifiers represented by the input. Note that to perform such checking for a known kind of entity, such as a constructor or method, first AND the argument of `toString` with the appropriate mask from a method like `constructorModifiers()` or `methodModifiers()`.

Parameters:

`mod - a set of modifiers`

Returns:

`a string representation of the set of modifiers represented by mod`

classModifiers

```
public static int classModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to a class.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a class.

Since:

1.7

See *The Java™ Language Specification*:

8.1.1 Class Modifiers

interfaceModifiers

```
public static int interfaceModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to an interface.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to an interface.

Since:

1.7

See *The Java™ Language Specification*:

9.1.1 Interface Modifiers

constructorModifiers

```
public static int constructorModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to a constructor.

Returns:

an `int` value OR-ing together the source language modifiers that can be applied to a constructor.

Since:

1.7

See *The Java™ Language Specification*:

8.8.3 Constructor Modifiers

methodModifiers

```
public static int methodModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to a method.

Returns:

an int value OR-ing together the source language modifiers that can be applied to a method.

Since:

1.7

See *The Java™ Language Specification*:

8.4.3 Method Modifiers

fieldModifiers

```
public static int fieldModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to a field.

Returns:

an int value OR-ing together the source language modifiers that can be applied to a field.

Since:

1.7

See *The Java™ Language Specification*:

8.3.1 Field Modifiers

parameterModifiers

```
public static int parameterModifiers()
```

Return an `int` value OR-ing together the source language modifiers that can be applied to a parameter.

Returns:

an int value OR-ing together the source language modifiers that can be applied to a parameter.

Since:

1.8

See *The Java™ Language Specification*:

8.4.1 Formal Parameters

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